Instructor's Note:

Not part of the class project, and not extra credit. ThermalCascade is one of three rate laws that ChatGPT 4o-mini proposed when asked for new reactive burn models; the report prints them as the model wrote them, with no human-generated content [1]. As written it cannot run. This sheet gives the two repairs that make it run; it is the second burn model in bitwhiz, the course's open-source hydrocode (`burn = thermalcascade`), for anyone to use. Find it, run it beside HVRB, and see what a temperature-driven rate does differently.

These equations build upon U3. Continuum Mechanics & Balance Laws, the Class Project: Hydrocode sheets (Method AV and Method RS), Class Project: Reactive Burn, HVRB (its products and closure), and bitwhiz: Reactive Burn, the Lead Shock

What ThermalCascade Adds to Your Hydrocode:

- The HVRB sheet's progress variable, $\boldsymbol{\lambda}$, in every Material X cell: $\boldsymbol{\lambda = 0}$ unreacted, $\boldsymbol{\lambda = 1}$ fully reacted
- A rate law, temperature and lead shock $\rightarrow \boldsymbol{\lambda}$ page 2
- The HVRB sheet's products and closure, unchanged, $\boldsymbol{(\rho,\ e,\ \lambda) \rightarrow P,\ T}$ here $\boldsymbol{T}$ drives the rate; it is no longer only an intermediate result
- A lead shock in every cell, its pressure $\boldsymbol{P_S}$, found from the cell's own history and then frozen; nothing burns before it the Lead Shock sheet
- Notation, rate coefficient $\boldsymbol{H_T}$, critical temperature $\boldsymbol{T_C}$, exponents $\boldsymbol{G_T}$ and $\boldsymbol{N_S}$, reference pressure $\boldsymbol{P_R}$ (HVRB's); time level $\boldsymbol{n}$
- $\boldsymbol{G_T}$ is not the products' $\boldsymbol{\gamma}$, and $\boldsymbol{H_T}$ is not U3's areal mass coordinate $\boldsymbol{h}$: the report's letters, renamed (next box)
- ***Everything else in your cycle is unchanged; the lead shock adds one check per cell per cycle***

Rate Law — as Proposed, and as Repaired [1]:

- As proposed, $\dfrac{d\lambda}{dt} = h\,(1-\lambda)\,(T-T_c)^{\gamma}\left(1+\dfrac{P_S}{P_{ref}}\right)^{p}$ verbatim, Sec. 5.2.3
- Renamed, $h \to H_T$ ("a heat transfer coefficient" there; here in $1/(\mu\text{s K}^{G_T})$), $\gamma \to G_T$, $p \to N_S$, $T_c \to T_C$, $P_{ref} \to P_R$
- Repair 1, $(T-T_C)^{G_T} \to [\max(T-T_C,\,0)]^{G_T}$ below T_C a non-integer power is not real, and a cell there must not burn
- Repair 2, the gate: $D\lambda/Dt = 0$ until the cell's lead shock is found, then P_S is frozen the report never says how P_S is found
- As repaired, $\dfrac{D\lambda}{Dt} = H_T\,(1-\lambda)\,[\max(T-T_C,\,0)]^{G_T}\left(1+\dfrac{P_S}{P_R}\right)^{N_S}$
- First order in $1-\lambda$, $g(\lambda) = 1-\lambda$ is largest at $\lambda = 0$, unlike HVRB's $g(0) = 0$: the delay behind the shock comes from T instead
- $G_T = 0$, read $[\,\cdot\,]^0$ as 1 above T_C and 0 at or below it: a clock, once the lead shock is found
- ***The cascade: burning heats the cell, and a hotter cell burns faster***

References:

[1] D. E. Kittell, A. R. Garcia, M. L. Hobbs, W. W. Erikson, K. P. Ruggirello, J. A. Brown, D. L. Damm, J. A. Stewart, P. A. Sable, J. W. LaJeunesse, E. N. Harstad, L. W. Tuttle, and A. L. Brundage, "A Short Survey of Current Reactive Burn Model Capabilities as of Mid-Year 2025," Sandia National Laboratories, SAND2025-09228 (July 2025), doi:10.2172/2585504.